

Clinical Research Article

Associations of Age at Menopause With Postmenopausal Bone Mineral Density and Fracture Risk in Women

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Abstract

Context: Menopause before age 45 is a risk factor for fractures, but menopause occurs at age ≥ 45 in ~90% of women.

Objective: To determine, in women with menopause at age ≥ 45 , whether (1) years since the final menstrual period (FMP) is more strongly associated with postmenopausal bone mineral density (BMD) than chronological age and (2) lower age at FMP is related to more fractures.

Design and Setting: The Study of Women's Health Across the Nation, a longitudinal cohort study of the menopause transition (MT).

Participants: A diverse cohort of ambulatory women (pre- or early perimenopausal at baseline, with 15 near-annual follow-up assessments).

Main Outcome Measures: Postmenopausal lumbar spine (LS) or femoral neck (FN) BMD ($n = 1038$) and time to fracture ($n = 1554$).

Results: Adjusted for age, body mass index (BMI), cigarette use, alcohol intake, baseline LS or FN BMD, baseline MT stage, and study site using multivariable linear regression, each additional year after the FMP was associated with 0.006 g/cm^2 ($P < 0.0001$) and 0.004 g/cm^2 ($P < 0.0001$) lower postmenopausal LS and FN BMD, respectively. Age was not related to FN BMD independent of years since FMP. In Cox proportional hazards regression, accounting for race/ethnicity, BMI, cigarette use, alcohol intake, prior fracture, diabetes status, exposure to bone-modifying medications/supplements, and study site,

the hazard for incident fracture was 5% greater for each 1-year decrement in age at FMP ($P = 0.02$).

Conclusions: Years since the FMP is more strongly associated with postmenopausal BMD than chronological age, and earlier menopause is associated with more fractures.

Key Words: menopause, bone mineral density, fracture, general population studies

The number of years elapsed since the final menstrual period (FMP) has been proposed as a metric for biological aging in women, because the menopause transition (MT) is accompanied by physiologic changes that are strongly linked to adverse health outcomes in later life (1,2). A corollary of this hypothesis is that earlier menopause is associated with worse health in older ages. Bone is an ideal system for testing this hypothesis and its corollary, because bone loss in women is intimately tied to reduced ovarian function (3-14). Bone mineral density (BMD) decline in women accelerates 1 year before the FMP, continues at a rapid pace for 3 years, and persists in postmenopause more slowly. Over the 5 years bracketing the FMP, total bone loss can exceed 10% (14) and is accompanied by alterations in bone quality that diminish fracture resistance (15-17). Indeed, the incidence of distal appendicular fractures increases during the first postmenopausal decade (18), foreshadowing future vertebral and hip fractures (19). Menopause is, accordingly, cited as a reason that women have greater prevalence of osteoporosis and more fractures than men (20).

Because BMD decline accelerates earlier in women whose FMP occurs earlier, many investigators have hypothesized that lower age at FMP leads to lower postmenopausal BMD at a given chronological age and greater fracture risk (21-38). In support of this thesis, various studies have demonstrated that primary ovarian insufficiency [menopause before age 40 (21,22)] and menopause before age 45 (23-28) are related to lower BMD and/or more fractures. Indeed, in clinical practice, menopause before age 45 is commonly recognized as a risk factor for fracture (39). However, approximately 90% of women have their FMP at or after age 45 years (40), and the association of age at FMP with subsequent BMD and fracture risk in this large majority of women is less certain (22,29). Such an association, if it exists, could be used to improve the prediction of BMD and fracture in all postmenopausal women.

The objective of this study was, therefore, to fill this knowledge gap and address the following 2 research questions in women with FMPs at age ≥ 45 . First, in postmenopausal women, does the number of years elapsed since the FMP predict lower BMD, independent of chronological age? If it does, then in women of the same chronological age, those with lower age at FMP will have lower BMD. Second, is a lower age at FMP associated with greater rates of fracture?

We used previously collected reproductive, BMD, and fracture data from the Study of Women's Health Across the Nation (SWAN), a longitudinal study of the MT in a diverse cohort of ambulatory women.

Methods

SWAN is a multicenter, longitudinal study of the MT in a diverse cohort of community-dwelling women (41). At study inception (1996), participants were between 42 and 52 years of age, were in premenopause (regular menstrual bleeding in the past year) or early perimenopause (less predictable menstrual bleeding at least once every 3 months), had an intact uterus with >1 ovary, and were not taking sex steroid hormones. The SWAN cohort included 3302 participants recruited from 7 US sites (Boston, MA; Chicago, IL; Detroit, MI; Pittsburgh, PA; Los Angeles, CA; Newark, NJ; Oakland CA). The SWAN Bone Cohort was a subset of 2407 participants from 5 sites (excluding Chicago and Newark, where bone assessments were not performed). To date, a total of 16 visits (1 baseline and 15 follow-up) spanning 2 decades have been conducted. Each SWAN clinical site obtained Institutional Review Board approval, and all participants provided written informed consent.

Study Sample Derivation

This study consisted of 2 analytic samples. We used a fracture analysis sample to test whether earlier age at FMP (age ≥ 45) is related to greater incidence of subsequent fracture. This sample was derived from the full SWAN cohort, of whom 1595 underwent a natural (nonsurgical) menopause at age ≥ 45 and had a known FMP date. From these women, we then excluded those who started a bone-beneficial or bone-detrimental medication prior to SWAN baseline ($n = 41$). Accordingly, the fracture analysis sample included 1554 women. Bone-beneficial medications included hormone therapy, calcitonin, calcitriol, bisphosphonates, denosumab, and parathyroid hormone. Bone-detrimental medications were oral or injectable glucocorticoids, aromatase inhibitors, gonadotropin-releasing hormone agonists, or antiepileptic medications.

To examine whether the number of years elapsed since the FMP (age ≥ 45) is associated with lower postmenopausal

BMD, independent of chronological age, we used a BMD analysis sample. This sample was derived from the subset of SWAN participants who were in the Bone Cohort. Again, starting with the 1595 participants who had a known FMP date at age ≥ 45 years, we excluded those who were not from a SWAN Bone site ($n = 404$), did not have a BMD measurement at least 3 years after their FMP ($n = 121$), or started a bone-beneficial or bone-detrimental medication before the SWAN baseline visit ($n = 32$). This left us with a BMD analysis sample of 1038 women.

Outcomes

Bone mineral density

For our first analysis, the outcome was BMD at the last SWAN visit (up to follow-up visit 15). BMD at the lumbar spine (LS) and femoral neck (FN) BMD (g/cm^2) were measured using dual-energy X-ray absorptiometry at each study visit. An anthropomorphic spine phantom was circulated for cross-site calibration. The Boston, Detroit, and Los Angeles sites began SWAN with Hologic 4500A models and then upgraded to Hologic Discover A hardware. Davis and Pittsburgh started SWAN with Hologic 2000 models and later upgraded to Hologic 4500A machines. When a site upgraded hardware, it scanned 40 women on the old and new machines to develop cross-calibration regression equations. A standard quality control program included daily phantom measurements, local site review of all scans, central review of scans that met problem-flagging criteria, and central review of a 5% random sample of scans. Short-term in vivo measurement variability was $0.014 \text{ g}/\text{cm}^2$ (1.4%) at the LS and $0.016 \text{ g}/\text{cm}^2$ (2.2%) at the FN.

Fracture

Incident fracture was the outcome in our second analysis. Fractures were self-reported by all SWAN participants (even if not in the SWAN Bone Cohort) using standardized questionnaires. Questionnaires were administered at the SWAN baseline visit (to ascertain fractures prior to SWAN inception) and at each follow-up visit (to ascertain fractures that occurred since the previous visit after the start of SWAN). The site of all fractures was recorded. For fractures before SWAN baseline, age at the time of the fracture was elicited. For fractures during study follow-up, SWAN did not collect event dates during the first 6 follow-up visits; dates were thus imputed as the midpoint between the participants' previous and index visits. SWAN began collecting the date of fracture at visit 7. Also starting at visit 7, medical records were obtained to adjudicate fractures; since inauguration of adjudication, 95% of self-reports were confirmed. For fractures that occurred during SWAN, mechanism of injury was assessed. Fractures were

considered minimum trauma if they did not occur after a fall from a height of ≥ 6 inches, a motor vehicle accident, moving fast (eg, skating), playing sports, or from impact with heavy or fast-moving projectiles. Our analyses did not include craniofacial and digital fractures. However, we included minimum- and nonminimum-trauma fractures as both fracture types are associated with low BMD (42) and mechanism of injury was not recorded for fractures that occurred prior to SWAN baseline.

Primary Exposure: Age at the Final Menstrual Period

Participants were classified as postmenopausal after >1 year of amenorrhea. In women who underwent natural menopause, FMP date was defined as the first date of the last menstrual bleeding cycle before 12 months of amenorrhea.

Covariates

Risk factors for low BMD and fracture were included as covariates in analyses, as appropriate. These included age (years), race/ethnicity, body mass index [BMI; calculated as $\text{weight in kilograms}/(\text{height in meters})^2$], cigarette use (current/past/never), alcohol intake (<7 drinks per week, >7 and <14 drinks per week, and >14 drinks per week), prior fracture (yes/no), diabetes status (yes/no), and use of bone-beneficial or bone-detrimental medications and/or vitamin D or calcium supplementation. Each of these variables was collected at every SWAN visit using standardized self-report or interview forms. Anthropometrics were ascertained using standardized and quality-controlled protocols (41).

Statistical Analysis

Descriptive statistics for all variables were generated and distributions of continuous variables were assessed for normality.

Our first analysis tested whether time elapsed since the FMP is associated with lower postmenopausal BMD, independent of chronological age. At any given chronological age in postmenopause, women who had earlier FMPs have had more years elapsed since the FMP; thus, this analysis also tests the hypothesis that at a given chronological age, earlier FMP is associated with lower BMD. We used multivariable linear regression to model BMD at the last SWAN visit as a function of years elapsed since the FMP and adjusted for chronological age (at the time of the outcome BMD assessment). Additional nontime-varying covariates included race/ethnicity, SWAN study site, and exposure to bone-beneficial or bone-detrimental

medications and/or vitamin D or calcium supplementation. Time-varying covariates assessed at SWAN baseline were BMI (kg/cm^2), cigarette use (current/past/never), alcohol intake (<7 drinks per week, >7 and <14 drinks per week, and >14 drinks per week), LS or FN BMD, and MT stage (pre- or early perimenopause). Exposure to bone-beneficial or bone-detrimental medications and/or vitamin D or calcium supplementation was operationalized as the percentage of visits from SWAN study baseline to the last SWAN visit that the participant reported taking the medication or supplement in question. Separate models were run for the LS and FN.

Our second analysis examined whether lower age at FMP is associated with greater fracture hazard. We used Cox proportional hazards regression with time to first fracture as the outcome and age at FMP as the continuous primary predictor. We started the observation period clock for fracture at age 40 to address potential confounding by chronological age and because MT-related BMD loss begins prior to the FMP (14), and we did not want to miss fractures that may have occurred before the FMP. Covariates included race/ethnicity, SWAN study site, BMI (kg/cm^2), history of fracture before age 40 (yes/no), cigarette use (current/past/never), alcohol intake (<7 drinks per week, >7 and <14 drinks per week, and >14 drinks per week), diabetes status (yes/no), exposure to bone-beneficial or bone-detrimental medications and/or vitamin D or calcium supplementation. Values for BMI, cigarette use, alcohol intake, and diabetes status were obtained at study baseline. Exposure to bone-beneficial or bone-detrimental medications and/or vitamin D or calcium supplementation was modeled as the percentage of visits from age 40 to the time of censoring or fracture that the participant reported taking the drug in question. To examine whether age at FMP is related to incident fracture independent of BMD in pre- and early perimenopause, we used participants in the fracture analysis sample who were recruited from Bone Cohort sites and ran the same Cox proportional hazards regression model as previously described but additionally adjusted for LS or FN BMD plus MT stage (pre- or early perimenopause) at the SWAN baseline visit.

Results

Participant Characteristics

Analytic sample characteristics are summarized in Table 1. The BMD analysis sample consisted of 1038 participants who had a known FMP date at age ≥ 45 and a BMD assessment at least 3 years after the FMP. Twenty-seven percent of participants were black, 13% Chinese, 15% Japanese, and 45% white. Mean age at FMP was 52 years. At the time of the last BMD measurement, participants were, on

average, 61.9 years of age, and had undergone 13 BMD assessments.

The fracture analysis sample included 1554 women who had a known FMP date at age ≥ 45 and did not use a bone-beneficial or bone-detrimental medication before the SWAN baseline visit. The racial/ethnic composition in this sample was 29% black, 9% Chinese, 7% Hispanic, 11% Japanese, and 44% white. On average, the FMP occurred at age 52 years in this sample.

Age at FMP and Bone Mineral Density

In the BMD analysis sample, mean (SD) LS and FN BMD at the last SWAN follow-up visit was 0.957 (0.170) and 0.739 (0.131) g/cm^2 , respectively. In postmenopausal women who underwent menopause at age ≥ 45 , greater time elapsed since the FMP (ie, lower age at the FMP) was associated with lower BMD, independent of chronological age. In multivariable linear regression, adjusted for chronological age, BMI, cigarette use, alcohol intake, exposure to bone-beneficial or bone-detrimental medications and/or vitamin D or calcium supplementation, baseline BMD, baseline MT stage, and study site, each additional year after the FMP was associated with 0.006 ($P < 0.0001$) and 0.004 ($P = 0.0001$) g/cm^2 lower LS and FN BMD, respectively, at the last SWAN visit. Thus, compared to women with menopause at age 55, women with menopause at age 47 had 0.048 and 0.032 g/cm^2 lower BMD at the LS and FN, respectively. Along these lines, LS and FN BMD were 0.030 and 0.020 g/cm^2 lower, respectively, in those with FMP at age 47 vs FMP at 52 years. Adjusted for time from FMP, chronological age was not associated with BMD at the FN, but greater chronological age was related to greater BMD at the LS. Years elapsed since the FMP and baseline LS or FN BMD were independently associated with postmenopausal BMD. Controlling for baseline BMD, race/ethnicity was not associated with LS or FN BMD at the last SWAN visit (Table 2).

Age at FMP and Incident Fracture

In the fracture analysis sample, a total of 317 women sustained an incident fracture. Fractures occurred most frequently at distal appendicular sites, with ankle, wrist, and foot being the most common. Mean duration of the observation period was 22 years, meaning that, on average, the analysis followed women into their 60s.

In women with natural menopause at age ≥ 45 , menopause at a lower age was associated with more future fractures. Using Cox proportional hazards regression and accounting for race/ethnicity, BMI, cigarette use, alcohol intake, prior fracture, diabetes status, exposure to

Table 1. Descriptive statistics for analytic samples

	BMD sample ^a (n = 1038)	Fracture analysis sample ^b (n = 1554)
Age at final menstrual period, years	52.1 (2.7)	52.1 (2.8)
Race/ethnicity:		
Black	278 (27)	443 (29)
Chinese	136 (13)	147 (9)
Hispanic	-	105 (7)
Japanese	159 (15)	178 (11)
White	465 (45)	681 (44)
At SWAN baseline		
Chronological age, years	46.5 (2.7)	46.5 (2.7)
Body mass index, kg/m ²	27.2 (6.8)	28.1 (7.3)
Lumbar spine BMD, g/cm ²	1.07 (0.14)	-
Femoral neck BMD, g/cm ²	0.84 (0.13)	-
Cigarette use		
Never	629 (61)	907 (58)
Past	255 (25)	386 (25)
Current	154 (14)	261 (17)
Alcohol intake:		
0 per week	560 (54)	807 (52)
<7 per week	402 (439)	634 (41)
>7 to 14 per week	51 (5)	78 (5)
>14 per week	25 (2)	35 (2)
Diabetes, yes	39 (4)	76 (5)
At last SWAN visit		
Chronological age, years	61.9 (4.2)	-
Time from final menstrual period, years	9.1 (4.0)	-
Lumbar spine BMD, g/cm ²	0.957 (0.170)	-
Femoral Neck BMD, g/cm ²	0.739 (0.131)	-

Data are given as n (%) for categorical variables and mean (SD) for continuous variables.

Abbreviations: BMD, bone mineral density; SWAN, Study of Women's Health Across the Nation.

^aBMD analysis sample is used to examine the association of age at final menstrual period (FMP) with future BMD. Sample includes women who had a natural menopause and known FMP date at age 45 and later, did not start a bone-beneficial or bone-detrimental medication before the SWAN baseline visit, and had a BMD measurement at least 3 years after the FMP. Bone-beneficial medications included hormone therapy, calcitonin, calcitriol, bisphosphonates, denosumab, and parathyroid hormone. Bone-detrimental medications included oral or injectable glucocorticoids, aromatase inhibitors, gonadotropin releasing-hormone agonists, or antiepileptic medications.

^bFracture analysis sample is used to examine the relation of age at final menstrual period (FMP) with subsequent fracture. Sample includes women who had a natural menopause and known FMP date at age 45 and later and did not start a bone-beneficial or bone-detrimental medication before the SWAN baseline visit. Bone-beneficial medications included hormone therapy, calcitonin, calcitriol, bisphosphonates, denosumab, and parathyroid hormone. Bone-detrimental medications included oral or injectable glucocorticoids, aromatase inhibitors, gonadotropin-releasing hormone agonists, or antiepileptic medications.

bone-beneficial or bone-detrimental medications and/or vitamin D or calcium, and study site, the hazard for incident fracture was 5% greater for each 1-year decrement in age at FMP ($P = 0.02$) (Table 3). Thus, the hazard for fracture was 23% and 34% greater in women with menopause at age 47 vs menopause at ages 52 and 55, respectively. In terms of fracture risk, the estimated 20-year cumulative fracture probability in women with FMP at age 47 was 4.1% greater than in women with FMP at age 52 (20.2% vs 16.1%) and 6.3% greater than in women with FMP at age 55 (20.2% vs 13.9%). A diagnosis of diabetes and prior fracture were related to more future fractures. Compared to white women, black and Japanese women had lower rates of fractures.

Lastly, we examined whether FMP at a lower age was associated with subsequent fracture, independent of baseline BMD using a subset of 1146 women from the fracture analysis sample recruited from Bone Cohort sites. Of these volunteers, 224 sustained an incident fracture. In Cox proportional hazards regression adjusted for the same covariates as previously listed, plus baseline LS or FN BMD and baseline MT stage, each 1-year decrement in age at FMP was associated with a 5% greater hazard for incident fracture hazard (hazard ratio = 0.95, $P = 0.04$). Higher starting LS and FN BMD were associated with lower rates of subsequent fracture, independent of age at FMP. Each SD greater BMD at the LS and FN was associated with 33% and 31% lower fracture hazard ($P < 0.0001$), respectively.

Table 2. Adjusted associations of time elapsed after the final menstrual period and key covariates with postmenopausal bone mineral density

	BMD (g/cm ²) at the last study visit			
	Lumbar spine	P-value	Femoral neck	P-value
Time from FMP, years	−0.006 (−0.08, −0.004)	<0.0001	−0.004 (−0.006, −0.002)	<0.0001
Chronological age, years	0.006 (0.003, 0.09)	<0.0001	−0.000 (−0.002, 0.002)	0.7
BMI, kg/m ²	0.004 (0.009, 0.012)	<0.0001	0.000 (−0.001, 0.001)	0.5
Race/ethnicity				
Black	−0.005 (−0.020, 0.011)	0.5	−0.004 (−0.019, 0.011)	0.4
Chinese	0.001 (−0.026, 0.027)	0.9	−0.003 (−0.023, 0.018)	0.7
Japanese	0.002 (−0.024, 0.027)	0.9	−0.012 (−0.031, 0.008)	0.2
White	Reference		Reference	
Starting BMD				
Lumbar spine, SD	0.166 (0.159, 0.174)	<0.0001	—	—
Femoral neck, SD	—	—	0.129 (0.122, 0.136)	<0.0001

Associations are presented as point estimate (95% CI) and are results of multivariable linear regression with the last postmenopausal lumbar spine or femoral neck bone mineral density (BMD) measurement in each participant as outcome and the number of years from the final menstrual period at the time of the BMD measurement as primary predictor. Models were adjusted for chronological age at the time of the BMD assessment, race/ethnicity, Study of Women's Health Across the Nation (SWAN) study site, exposure to bone-beneficial or bone-detrimental medications and/or vitamin D or calcium supplementation, and the following covariates assessed at SWAN baseline: body mass index (kg/cm²), cigarette use (current/past/never), and alcohol intake (<7 drinks per week, >7 and <14 drinks per week, and >14 drinks per week), lumbar spine or femoral neck BMD, and menopause transition stage (pre- or early perimenopause). Bone-beneficial medications included hormone therapy, calcitonin, calcitriol, bisphosphonates, denosumab, and parathyroid hormone. Bone-detrimental medications included oral or injectable glucocorticoids, aromatase inhibitors, gonadotropin-releasing hormone agonists, or antiepileptic medications.

Abbreviations: BMD, bone mineral density; BMI, body mass index; FMP, final menstrual period.

Table 3. Associations of age at the final menstrual period and key covariates with fracture risk

	Hazard ratio for incident fracture (95% CI)	P-value
Age at FMP, years	0.95 (0.91, 0.99)	0.02
Race/ethnicity		
Black	0.55 (0.46, 0.75)	0.0002
Chinese	0.73 (0.41, 1.31)	0.2
Hispanic	0.50 (0.22, 1.14)	0.1
Japanese	0.47 (0.27, 0.84)	0.01
White	Reference	
Fracture before age 40, yes/no	1.62 (1.17, 2.23)	0.003
Diabetes status, yes/no	1.81 (1.14, 2.89)	0.01

Associations were determined by Cox proportional hazards regression with time to first fracture as the outcome and age at FMP as primary predictor. The model was adjusted for race/ethnicity, BMI, history of fracture before age 40, cigarette use, alcohol intake, diabetes status, exposure to bone-beneficial or bone-detrimental medications and/or vitamin D or calcium supplementation, and study site. Bone-beneficial medications included hormone therapy, calcitonin, calcitriol, bisphosphonates, denosumab, and parathyroid hormone. Bone-detrimental medications included oral or injectable glucocorticoids, aromatase inhibitors, gonadotropin-releasing hormone agonists, or anti-epileptic medications.

Abbreviation: FMP, final menstrual period.

Discussion

The objectives of this study were to examine whether the number of years since the FMP (in women whose FMP occurs at age ≥45) is more strongly associated with postmenopausal BMD than chronological age and whether earlier menopause is related to more fractures. Our analyses demonstrate that among postmenopausal women, the number of years elapsed since the FMP is more strongly associated with LS and FN BMD than chronological age.

In fact, chronological age did not predict BMD at the FN after accounting for years since the FMP and was associated with greater BMD at the LS. The unexpected positive association between chronological age and LS BMD (for a given number of years since the FMP) is likely a reflection of age-associated degenerative changes causing false elevations in BMD measured by dual-energy X-ray absorptiometry (43). Our analysis also demonstrated that lower age at FMP is associated with greater incidence of fractures.

Menopause before age 45 is commonly considered a risk factor for lower BMD and fractures (21–28), but the relation between age at FMP with subsequent BMD and fractures in women who have their FMP at age ≥ 45 is less certain (29). For example, the Women's Health Initiative published that women with primary ovarian insufficiency (FMP before age 40) had lower BMD and a 21% greater fracture hazard compared to women whose FMP occurred after age 50 (21). More recently, a meta-analysis of over 460 000 women reported that menopause before age 45 was associated with a 36% greater odds of fracture (28).

Our study contributes to the current body of literature by specifically examining the relation between age at FMP with subsequent BMD and fracture in women whose FMPs occurred at age ≥ 45 . This population is important to study because ~90% of women transition to postmenopause after age 45 (40). Indeed, in this group of women, the gradient of risk between age at FMP with fracture was substantial: the absolute difference in 20-year cumulative fracture probability was 6.3% greater for women with menopause at age 47 vs 55 years. Our results are consistent with findings from the Malmo Perimenopausal Study, which revealed that menopause before age 47 was associated with an 83% and 59% greater risk of densitometric osteoporosis and fracture, respectively, by age 77 (29). In contrast to the Malmo study, we only included women with an age at FMP of ≥ 45 so that women with menopause before age 45 would not drive our results.

Physiologically, earlier FMP means earlier MT-related acceleration in BMD loss and worse bone outcomes in later life. BMD decline accelerates approximately 1 year before the FMP, decreases at its greatest rate during the MT (1 year before to 2 years after the FMP), and persists at a lower rate in postmenopause (14). In turn, rapid MT-related and postmenopausal bone loss is accompanied by declines in bone quality and strength (15–17). It is, therefore, not surprising that earlier menopause means lower BMD at a given chronological age and more fractures. These results add to a growing body of literature suggesting that the endocrine changes that occur during the MT trigger a pathophysiologic cascade that leads to organ dysfunction (44). Within this construct, earlier menopause has been proposed as a marker for more advanced biological age (2). To date, adverse cardiovascular events and outcomes, frailty, and even shorter life expectancy have all been linked to lower age at FMP (1–13).

Clinically, knowing that time since the FMP is more strongly related to postmenopausal BMD than chronological age and that earlier menopause is a risk factor for fractures may ultimately aid prediction of these outcomes. Future studies should examine whether years since the FMP predicts major osteoporotic fractures and hip fractures, specifically and, if so, whether replacing chronological age with years since the FMP improves the performance of clinical prediction tools, such as FRAX (39).

A methodologic strength of this study is that, in SWAN, age at FMP was collected during real-time, observational follow-up. This may be one reason that we were able to discern relations of age at FMP with BMD and fracture risk, whereas other studies that ascertained age at FMP by recall (often decades after participants transitioned to postmenopause) (34–38) could not. Our study has several limitations that warrant mention. First, while fractures of the spine and hip are associated with the greatest morbidity and mortality (45,46), the majority of fractures sustained in our younger study cohort occurred at distal appendicular sites. In addition, we did not have a sufficient number of fractures at each anatomic site to examine the association of age at FMP with fracture type. Despite these limitations, identifying risk factors for distal appendicular fractures is clinically important, as distal appendicular fractures are harbingers of future spine and hip fractures (47–52). Lastly, because the number of fracture events in nonwhite women was relatively small, we did not have sufficient power to formally test whether the associations of age at FMP with postmenopausal BMD and incident fracture were modified by race/ethnicity.

Conclusion

To summarize, this study demonstrates that in women who become postmenopausal at age ≥ 45 , lower age at FMP is associated with lower postmenopausal BMD and more future fractures. Future studies will examine whether combining age at FMP with other clinical and biochemical factors can improve risk stratification for future fractures.

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Additional Information

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